GEOMETRY · UNIT 10: AREA, SURFACE AREA AND VOLUME

Popcorn Container Challenge

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students compare a cylinder, a cone, and a box of popcorn by volume, surface area, and price per cubic inch, use pyramids, spheres, and scale factors, and design a container that holds the most popcorn with the least cardboard.

Volume and surface area become competing goals. Students see that shapes with the same height and base can hold very different amounts (cone vs. cylinder), and that scaling affects volume by k^3.

Standards Addressed

- G-GMD.3 Volume formulas for cylinders, pyramids, cones, and spheres
- G-MG.1 Use shapes to describe objects
- G-MG.3 Solve design problems
- G-GMD.1 Cavalieri's principle (Honors)

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch: which container holds the most? Guess, then test with beans. Part 1.
Day 2	Part 2 other solids and scale factors.
Day 3	Part 3 design and build.

Materials

- Student handout
- Cardstock, rulers, scissors, tape
- Popcorn or dried beans to test volume (optional)
- Calculators

How the Three Versions Differ

Level	What changes
CP2	Volumes and price per cubic inch of three containers, plus open-top surface area with formulas given.
CP1	Pyramids, spheres, lateral area with slant height, and scale factors.
Honors	Cavalieri's principle, optimization of an open cylinder with a fixed volume, and a design proof.

Watch For (Common Misconceptions)

- Forgetting the ⅓ in cone and pyramid volume.
- Including a lid in an open-top container's surface area.
- Using diameter instead of radius.

Extension Ideas

- Build the designs and test them with popcorn.
- Compare real theater containers.
- Graph surface area against radius for a fixed volume.

Answer Key – CP2

Part 1: Which Holds the Most?

#	Task	Answer
1	Find the volume of the cylinder ($V = \pi r^2 h$).	$\pi(4)^2(8) = 128\pi \approx 402.12$ in^3
2	Find the volume of the cone ($V = \frac{1}{3}\pi r^2 h$).	$\frac{1}{3}\pi(4)^2(12) = 64\pi \approx 201.06$ in^3
3	Find the volume of the box ($V = lwh$).	$6 \times 6 \times 10 = 360$ in^3
4	Rank the containers from most to least popcorn.	Cylinder, box, cone

Part 2: Value & Material

#	Task	Answer
5	Find the price per cubic inch for each container. Which is the best value?	Cylinder \$6 ÷ 402.12 ≈ \$0.0149; cone \$3.50 ÷ 201.06 ≈ \$0.0174; box \$5 ÷ 360 ≈ \$0.0139 per in^3. The box is the best value.
6	The cylinder has an open top. Its surface area is the bottom plus the side: $\pi r^2 + 2\pi rh$. Find it.	$16\pi + 64\pi = 80\pi \approx 251.33$ in^2
7	The box has an open top. Find its surface area (bottom plus four sides).	$36 + 4(60) = 276$ in^2
8	Which container uses less cardboard for each cubic inch of popcorn, the cylinder or the box?	Cylinder: 251.33 ÷ 402.12 ≈ 0.63 in^2 per in^3. Box: 276 ÷ 360 ≈ 0.77. The cylinder uses less.

Part 3: Design a Container

#	Task	Answer
9	Choose a shape and dimensions. Show that the volume is close to 300 in^3.	Answers vary.
10	Find the surface area (open top).	Answers vary.
11	Choose a price and find the price per cubic inch. Is it a good deal compared with the others?	Answers vary.

Answer Key – CP1

Part 1: Which Holds the Most?

#	Task	Answer
1	Find the volume of each container. Rank them.	Cylinder $128\pi \approx 402.12$; box 360; cone $64\pi \approx 201.06$ in^3. Cylinder, box, cone.
2	A new upside-down square pyramid container has an 8-in base and a 9-in height. Find its volume.	$\frac{1}{3} \times 64 \times 9 = 192$ in^3
3	Popcorn balls have a radius of 1.5 in. Find the volume of one ball. About how many would fill the cylinder by volume?	$(4 \div 3)\pi(1.5)^3 = 4.5\pi \approx 14.14$ in^3; $402.12 \div 14.14 \approx 28$ balls (fewer in practice because of gaps).
4	Find the cone's slant height and lateral (side) area.	$l = \sqrt{(4^2 + 12^2)} = \sqrt{160} \approx 12.65$ in; lateral area $\pi(4)(12.65) \approx 158.95$ in^2

Part 2: Scale & Value

#	Task	Answer
5	A "jumbo" cylinder doubles the radius but keeps the height. How many times as much popcorn does it hold?	4 times (radius squared)
6	A "mega" box is similar to the box with scale factor 1.5. Find its volume and how many times larger it is.	$1.5^3 = 3.375$ times: $360 \times 3.375 = 1{,}215$ in^3
7	The theater wants the mega box priced fairly (same price per cubic inch as the box). What should it cost?	$5 \times 3.375 = \$16.88$
8	The cylinder's label wraps around its side. Find the label's area.	$2\pi(4)(8) = 64\pi \approx 201.06$ in^2

Part 3: Design a Container

#	Task	Answer
9	Choose a shape and dimensions and show the volume is in range.	Answers vary.
10	Find the surface area and compare with the box.	Answers vary.
11	Sketch a net with dimensions.	Answers vary.

Answer Key – Honors

Part 1: Volumes & Cavalieri

#	Task	Answer
1	Find the exact volumes of all three containers.	Cylinder 128π, cone 64π, box 360 (in^3)
2	Explain with Cavalieri's principle why a slanted (oblique) cylinder with r = 4 and height 8 holds the same amount as the straight one.	Each horizontal cross-section is a circle with r = 4 at every height, so the volumes are equal.
3	A hemisphere bowl and a cone have the same radius r and the cone's height is r. Show the bowl holds twice as much.	Hemisphere $\frac{2}{3}\pi r^3$; cone $\frac{1}{3}\pi r^2 \cdot r = \frac{1}{3}\pi r^3$. The hemisphere is twice the cone.

Part 2: Optimize

#	Task	Answer
4	Write h in terms of r. Then write the surface area $S(r) = \pi r^2 + 2\pi rh$ in terms of r only.	$h = 400 \div (\pi r^2)$; $S(r) = \pi r^2 + 800 \div r$
5	Make a table of S(r) for r = 3, 4, 5, 6, 7. Which r gives the least cardboard?	r = 3: 294.9; 4: 250.3; 5: 238.5; 6: 246.4; 7: 268.2 (in^2). r = 5 is best among these.
6	The best design has r = h. Find this r to the nearest hundredth and the surface area.	$r = h = \sqrt[3]{(400 \div \pi)} \approx 5.03$ in; $S \approx 238.5$ in^2
7	How much cardboard does the best design save compared with the current cylinder (which holds 402 in^3)?	$251.3 - 238.5 \approx 12.8$ in^2 less (about 5%).

Part 3: Design & Prove

#	Task	Answer
8	Optimize one dimension using a table or graph.	Answers vary.
9	Compare your design with the optimal cylinder.	Answers vary.
10	Write a pitch that uses your math to persuade the manager.	Answers vary.

Scoring Rubric (all levels)

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Design meets the volume target and uses less material than at least one other container, with correct formulas. Goes beyond the minimum with a creative, realistic choice.	Design meets the volume target and uses less material than at least one other container, with correct formulas.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Pitch includes a net or model, volume and surface area, price per cubic inch, and a comparison with the other containers. Polished, organized, and visually engaging.	Pitch includes a net or model, volume and surface area, price per cubic inch, and a comparison with the other containers. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.